THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5	$\sqrt{3^2+4^2}=5$
2	**(b)** 3	Perimeter ratio = side ratio
3	**(c)** 6	$\frac{AD}{DB}=\frac{AE}{EC}$
4	**(d)** 3	Side ratio is the square root of the area ratio
5	**(a)** 5	Pythagoras
6	**(a)** equal	AAA similarity
7	**(a)** 13	$\sqrt{5^2+12^2}=13$
8	**(d)** 5	Perimeter ratio = side ratio
9	**(d)** 8	$\frac{AD}{DB}=\frac{AE}{EC}$
10	**(c)** 5	Side ratio is the square root of the area ratio

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$h^2=36+64=100$ (1); $h=10$ cm (1).	1+1
12	Area ratio $=(2/3)^2$ (1) $=4:9$ (1).	1+1
13	$\frac{AD}{DB}=\frac{AE}{EC}$ (1); $EC=8$ cm (1).	1+1
14	$h^2=81+144=225$ (1); $h=15$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$h^2=3^2+4^2=25$ (1); $h=5$ cm (1); perimeter $=12$ cm (1).	1+1+1
16	By the Basic Proportionality Theorem, $\frac{AD}{DB}=\frac{AE}{EC}$ (1); $\frac{2}{3}=\frac{4}{EC}$ (1); $EC=6$ cm (1).	1+1+1
17	Largest side 10: $(10)^2=100$ (1); sum of squares of the others $=100$ (1); equal, so by the converse of Pythagoras it is right-angled at the vertex opposite the longest side (1).	1+1+1
	OR: $h^2=5^2+12^2=169$ (1); $h=13$ cm (1); perimeter $=30$ cm (1).	1+1+1
18	By the Basic Proportionality Theorem, $\frac{AD}{DB}=\frac{AE}{EC}$ (1); $\frac{3}{5}=\frac{6}{EC}$ (1); $EC=10$ cm (1).	1+1+1
	OR: Largest side 9: $(9)^2=81$ (1); sum of squares of the others $=74$ (1); not equal, so it is not right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ladder is the hypotenuse of a right triangle (1); $d^2=5^2-4^2=9$ (1); $d=3$ m (1).	1+1+1
	OR: The triangles formed are similar (AA) (1); $\frac{H}{6}=\frac{30}{4}$ (1); $H=45$ m (1).	1+1+1
20	Ladder is the hypotenuse of a right triangle (1); $d^2=13^2-12^2=25$ (1); $d=5$ m (1).	1+1+1
	OR: The triangles formed are similar (AA) (1); $\frac{H}{5}=\frac{21}{3}$ (1); $H=35$ m (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB}=\frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC\sim\triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).
- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 13	Pythagoras
2	**(d)** $3^2 + 4^2 = 5^2$	Converse of Pythagoras
3	**(c)** 17	$\sqrt{8^2 + 15^2} = 17$
4	**(a)** 4	Perimeter ratio = side ratio
5	**(c)** 15	$\frac{AD}{DB} = \frac{AE}{EC}$
6	**(a)** 4	Side ratio is the square root of the area ratio
7	**(b)** 17	Pythagoras
8	**(d)** $5^2 + 12^2 = 13^2$	Converse of Pythagoras
9	**(b)** 25	$\sqrt{7^2 + 24^2} = 25$
10	**(c)** 7	Perimeter ratio = side ratio

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area ratio $= (3/4)^2$ (1) $= 9 : 16$ (1).	1+1
12	$\frac{AD}{DB} = \frac{AE}{EC}$ (1); $EC = 10$ cm (1).	1+1
13	$h^2 = 144 + 256 = 400$ (1); $h = 20$ cm (1).	1+1
14	Area ratio $= (5/2)^2$ (1) $= 25 : 4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$h^2 = 8^2 + 15^2 = 289$ (1); $h = 17$ cm (1); perimeter $= 40$ cm (1).	1+1+1
16	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{4}{5} = \frac{12}{EC}$ (1); $EC = 15$ cm (1).	1+1+1
17	Largest side 41: $(41)^2 = 1681$ (1); sum of squares of the others $= 1681$ (1); equal, so by the converse of Pythagoras it is right-angled at the vertex opposite the longest side (1).	1+1+1
	OR: $h^2 = 7^2 + 24^2 = 625$ (1); $h = 25$ cm (1); perimeter $= 56$ cm (1).	1+1+1
18	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{1}{4} = \frac{4}{EC}$ (1); $EC = 16$ cm (1).	1+1+1
	OR: Largest side 26: $(26)^2 = 676$ (1); sum of squares of the others $= 676$ (1); equal, so by the converse of Pythagoras it is right-angled at the vertex opposite the longest side (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ladder is the hypotenuse of a right triangle (1); $d^2 = 17^2 - 15^2 = 64$ (1); $d = 8$ m (1).	1+1+1
	OR: The triangles formed are similar (AA) (1); $\frac{H}{8} = \frac{35}{5}$ (1); $H = 56$ m (1).	1+1+1
20	Ladder is the hypotenuse of a right triangle (1); $d^2 = 25^2 - 24^2 = 49$ (1); $d = 7$ m (1).	1+1+1
	OR: The triangles formed are similar (AA) (1); $\frac{H}{4} = \frac{27}{3}$ (1); $H = 36$ m (1).	1+1+1

Mistakes to avoid:

- Writing the BPT as $\frac{AD}{AB} = \frac{AE}{EC}$; mix of a part-to-whole and a part-to-part ratio.
- Equating the *areas* ratio to the *sides* ratio instead of its square.
- Matching vertices in the wrong order while writing similarity ($\triangle ABC \sim \triangle PQR$ must have A to P, B to Q, C to R).
- Applying the Pythagoras theorem with the wrong side as hypotenuse (hypotenuse is opposite the right angle and is the longest side).
- Using the converse of Pythagoras without squaring all three sides, or comparing with the wrong side.
- Saying two triangles are similar because they have equal ratio of only two pairs of sides (SAS needs the included angle).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	$\frac{AD}{DB} = \frac{AE}{EC}$
2	**(b)** 5	Side ratio is the square root of the area ratio
3	**(b)** 25	Pythagoras
4	**(b)** $8^2 + 15^2 = 17^2$	Converse of Pythagoras
5	**(b)** 29	$\sqrt{20^2 + 21^2} = 29$
6	**(a)** 6	Perimeter ratio = side ratio
7	**(b)** 14	$\frac{AD}{DB} = \frac{AE}{EC}$
8	**(b)** 7	Side ratio is the square root of the area ratio
9	**(c)** 29	Pythagoras
10	**(d)** $7^2 + 24^2 = 25^2$	Converse of Pythagoras

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{AD}{DB} = \frac{AE}{EC}$ (1); $EC = 6$ cm (1).	1+1
12	$h^2 = 225 + 400 = 625$ (1); $h = 25$ cm (1).	1+1
13	Area ratio $= (4/7)^2$ (1) $= 16 : 49$ (1).	1+1
14	$\frac{AD}{DB} = \frac{AE}{EC}$ (1); $EC = 12$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$h^2 = 20^2 + 21^2 = 841$ (1); $h = 29$ cm (1); perimeter $= 70$ cm (1).	1+1+1
16	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{5}{2} = \frac{10}{EC}$ (1); $EC = 4$ cm (1).	1+1+1
17	Largest side 8: $(8)^2 = 64$ (1); sum of squares of the others $= 52$ (1); not equal, so it is not right-angled (1).	1+1+1
	OR: $h^2 = 9^2 + 40^2 = 1681$ (1); $h = 41$ cm (1); perimeter $= 90$ cm (1).	1+1+1
18	By the Basic Proportionality Theorem, $\frac{AD}{DB} = \frac{AE}{EC}$ (1); $\frac{3}{7} = \frac{9}{EC}$ (1); $EC = 21$ cm (1).	1+1+1
	OR: Largest side 37: $(37)^2 = 1369$ (1); sum of squares of the others $= 1369$ (1); equal, so by the converse of Pythagoras it is right-angled at the vertex opposite the longest side (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ladder is the hypotenuse of a right triangle (1); $d^2 = 15^2 - 12^2 = 81$ (1); $d = 9$ m (1).	1+1+1
	OR: The triangles formed are similar (AA) (1); $\frac{H}{7} = \frac{18}{2}$ (1); $H = 63$ m (1).	1+1+1
20	Ladder is the hypotenuse of a right triangle (1); $d^2 = 20^2 - 16^2 = 144$ (1); $d = 12$ m (1).	1+1+1
	OR: The triangles formed are similar (AA) (1); $\frac{H}{3} = \frac{20}{2}$ (1); $H = 30$ m (1).	1+1+1

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